

THE WHOLE OF HIGHER MATHS 2026 IN 5 HOURS!

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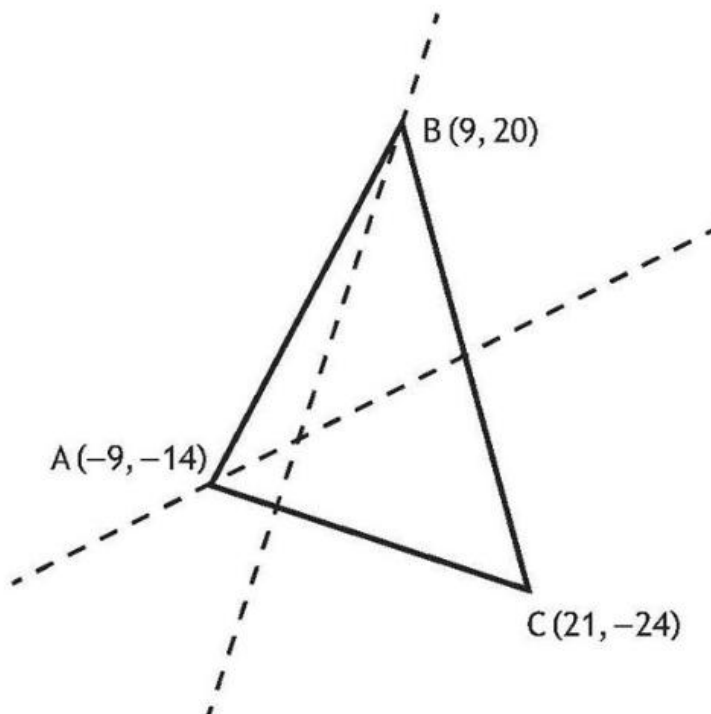
- STRAIGHT Line
- RECURRENCE RELATIONS
- FUNCTIONS AND GRAPHS
- TRIGONOMETRY
- DIFFERENTIATION
- EXPONENTIALS AND LOGS
- QUADRATICS AND POLYNOMIALS
- CIRCLES
- INTEGRATION
- VECTORS

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STRAIGHT LINE

Triangle ABC has vertices A $(-9, -14)$, B $(9, 20)$ and C $(21, -24)$.



- Find the equation of the altitude through B.
- Find the equation of the median through A.
- Determine the point of intersection of the altitude through B and the median through A.

Find the equation of the perpendicular bisector of the line joining A $(1, 4)$ and B $(9, 10)$.

The line, L , makes an angle of 30° with the positive direction of the x -axis.

Find the equation of the line perpendicular to L , passing through $(0, -4)$.

A, B and C are points such that AB is parallel to the line with equation $y + \sqrt{3}x = 0$ and BC makes an angle of 150° with the positive direction of the x -axis.

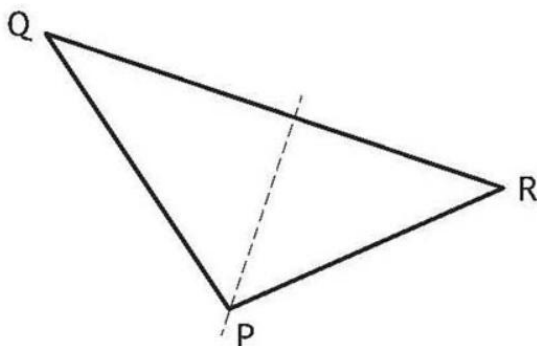
Are the points A, B and C collinear?

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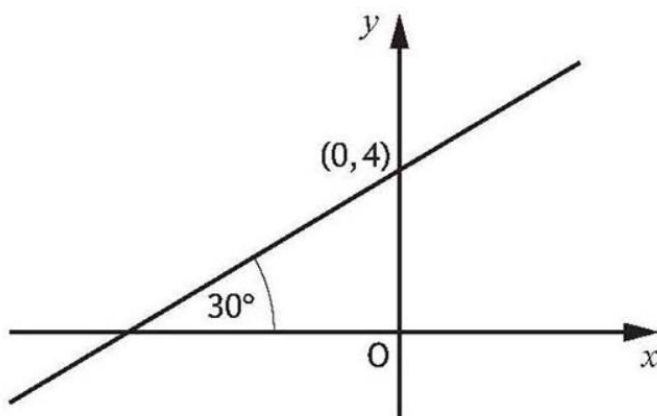
STRAIGHT LINE

Triangle PQR has vertices $P(5, -1)$, $Q(-2, 8)$ and $R(13, 3)$.



- Find the equation of the altitude from P.
- Calculate the angle that the side PR makes with the positive direction of the x -axis.

A line passes through the point $(0, 4)$ and makes an angle of 30° with the positive direction of the x -axis as shown in the diagram.



Determine the equation of the line.

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RECURRENCE RELATIONS

A sequence satisfies the recurrence relation $u_{n+1} = mu_n + 4$, where m is a constant.

(a)	The sequence approaches a limit of 10 as $n \rightarrow \infty$. Determine the value of m .	2
(b)	Given that $u_1 = 19$, calculate the value of u_0 .	1

A sequence is defined by the recurrence relation $u_{n+1} = \frac{1}{5}u_n + 12$ with $u_1 = 20$.

(a)	Calculate the value of u_2 .	1
(b)	(i) Explain why this sequence approaches a limit as $n \rightarrow \infty$.	1
	(ii) Calculate this limit.	2

In a forest, the population of a species of mouse is falling by 2.7% each year.
To increase the population scientists plan to release 30 mice into the forest at the end of March each year.

(a) u_n is the estimated population of mice at the start of April, n years after the population was first estimated.
It is known that u_n and u_{n+1} satisfy the recurrence relation $u_{n+1} = au_n + b$.
State the values of a and b . 1

The scientists continue to release this species of mouse each year.

(b)	(i) Explain why the estimated population of mice will stabilise in the long term.	1
	(ii) Calculate the long term population to the nearest hundred.	2

RECURRENCE RELATIONS

A sequence is generated by the recurrence relation

$$u_{n+1} = mu_n + c,$$

where the first three terms of the sequence are 6, 9 and 11.

- (a) Find the values of m and c . 3
- (b) Hence, calculate the fourth term of the sequence. 1

A version of the following problem first appeared in print in the 16th Century.

A frog and a toad fall to the bottom of a well that is 50 feet deep.
 Each day, the frog climbs 32 feet and then rests overnight. During the night, it slides down $\frac{2}{3}$ of its height above the floor of the well.
 The toad climbs 13 feet each day before resting.
 Overnight, it slides down $\frac{1}{4}$ of its height above the floor of the well.

Their progress can be modelled by the recurrence relations:

- $f_{n+1} = \frac{1}{3}f_n + 32, \quad f_1 = 32$
- $t_{n+1} = \frac{3}{4}t_n + 13, \quad t_1 = 13$

where f_n and t_n are the heights reached by the frog and the toad at the end of the n th day after falling in.

- (a) Calculate t_2 , the height of the toad at the end of the second day. 1
- (b) Determine whether or not either of them will eventually escape from the well. 5

FUNCTIONS & GRAPHS

A function, g , is defined by $g(x) = (x - 4)^3$, where $x \in \mathbb{R}$.

Find the inverse function, $g^{-1}(x)$.

3

Functions f and g are defined on $\mathbb{R}$, the set of real numbers, by:

- $f(x) = 2x^2 - 18$
- $g(x) = x + 1$.

(a) Find an expression for $f(g(x))$.

2

(b) Find the values of x for which $\frac{1}{f(g(x))}$ is undefined.

2

Functions f and g are defined on the set of real numbers by:

- $f(x) = x^5 + 3$
- $g(x) = 1 - x^3$.

(a) Find an expression for $h(x)$, where $h(x) = f(g(x))$.

2

(b) Find $h'(x)$.

2

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FUNCTIONS & GRAPHS

Functions f and g are defined by:

- $f(x) = 2 \sin x$, where $0 < x < \frac{\pi}{2}$
- $g(x) = 2x$, where $0 < x < \frac{\pi}{4}$

(a) (i) Evaluate $f\left(g\left(\frac{\pi}{6}\right)\right)$. 1

(ii) Determine an expression for $f(g(x))$. 2

(b) (i) Given that $f(p) = \frac{1}{3}$, determine the exact value of $\sin p$. 1

(ii) Hence, determine the exact value of $f(g(p))$. 3

A function, h , is defined by $h(x) = 2x^3 - 7$ where $x \in \mathbb{R}$.

Find the inverse function, $h^{-1}(x)$. 3

A function $f(x)$ is defined by $f(x) = \frac{2}{x} + 3$, $x > 0$.

Find the inverse function, $f^{-1}(x)$. 3

Functions f and g are defined on $\mathbb{R}$, the set of real numbers.

The inverse functions f^{-1} and g^{-1} both exist.

(a) Given $f(x) = 3x + 5$, find $f^{-1}(x)$. 3

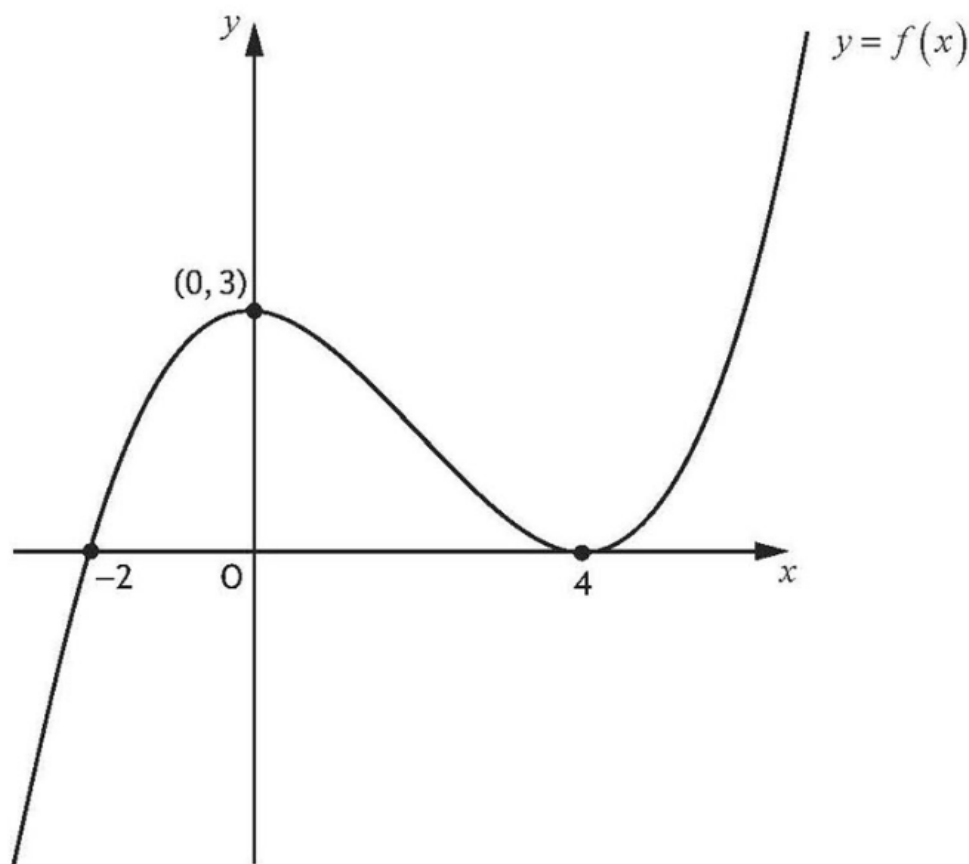
(b) If $g(2) = 7$, write down the value of $g^{-1}(7)$. 1

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FUNCTIONS & GRAPHS

The diagram shows the graph of $y = f(x)$, with stationary points at $(0, 3)$ and $(4, 0)$.



On the diagram in your answer booklet, sketch the graph of $y = f(-x) + 3$.

2

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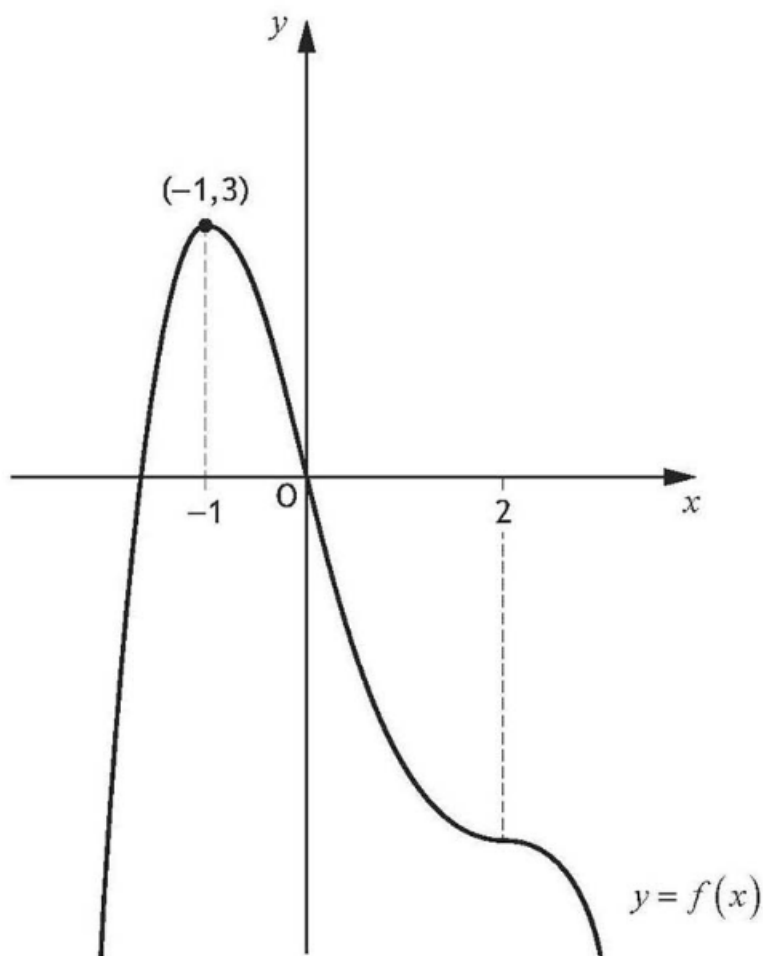
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FUNCTIONS & GRAPHS

The diagram shows the graph of a quartic function $y = f(x)$.

A maximum turning point occurs at $(-1, 3)$.

The graph of $y = f(x)$ also has a point of inflection at $x = 2$.



(a) Determine the coordinates of the maximum turning point on the graph of $y = f(x-4) + 2$.

2

(b) On the diagram in your answer booklet, sketch the graph of $y = f'(x)$.

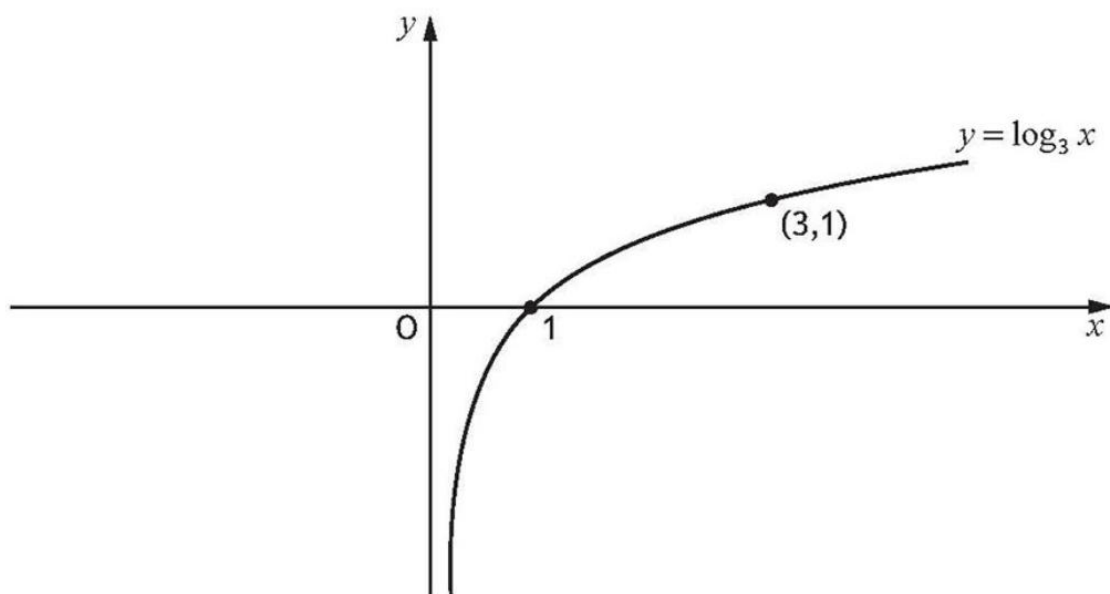
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FUNCTIONS & GRAPHS

The diagram shows the curve with equation $y = \log_3 x$.



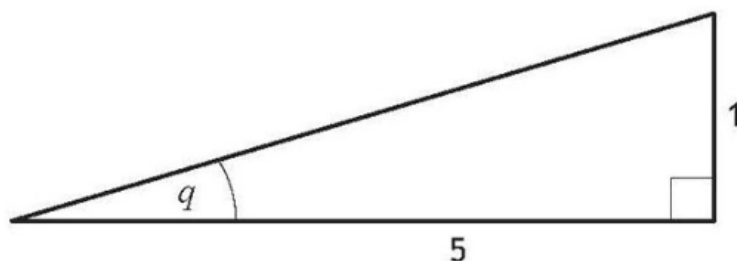
- (a) On the diagram in your answer booklet, sketch the curve with equation $y = 1 - \log_3 x$. 2
- (b) Determine the exact value of the x -coordinate of the point of intersection of the two curves. 3

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TRIGONOMETRY

The diagram shows a right-angled triangle with angle q .



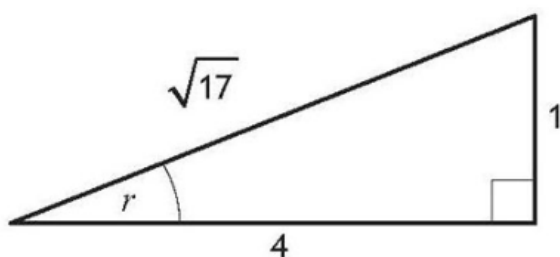
(a) Determine the value of:

- (i) $\sin 2q$
- (ii) $\cos 2q$.

3

1

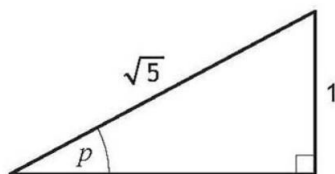
A second right-angled triangle has angle r as shown.



(b) Find the value of $\sin(2q - r)$.

3

The right-angled triangle in the diagram is such that $\sin p = \frac{1}{\sqrt{5}}$ and $0 < p < \frac{\pi}{4}$.



(a) Determine the value of:

- (i) $\sin 2p$
- (ii) $\cos 2p$.

3

1

(b) Hence determine the value of $\sin 4p$.

1

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TRIGONOMETRY

Given that $\tan 2x = \frac{3}{4}$, $0 < x < \frac{\pi}{4}$, find the exact value of

(a) $\cos 2x$ 1

(b) $\cos x$. 2

(a) Show that $\sin 2x \tan x = 1 - \cos 2x$, where $\frac{\pi}{2} < x < \frac{3\pi}{2}$. 4

(b) Given that $f(x) = \sin 2x \tan x$, find $f'(x)$. 2

(a) Show that $\frac{\sin 2x}{2 \cos x} - \sin x \cos^2 x = \sin^3 x$, where $0 < x < \frac{\pi}{2}$. 3

(b) Hence, differentiate $\frac{\sin 2x}{2 \cos x} - \sin x \cos^2 x$, where $0 < x < \frac{\pi}{2}$. 3

(a) Express $\cos x^\circ + \sqrt{3} \sin x^\circ$ in the form $k \cos(x - a)^\circ$, where $k > 0$ and $0 < a < 360$. 4

(b) Hence, or otherwise, sketch the graph with equation $y = \cos x^\circ + \sqrt{3} \sin x^\circ$, $0 \leq x \leq 360$. 3

Use the diagram provided in your answer booklet.

(a) Express $7 \cos x^\circ - 3 \sin x^\circ$ in the form $k \sin(x + a)^\circ$ where $k > 0$, $0 < a < 360$. 4

(b) Hence, or otherwise, find:

(i) the maximum value of $14 \cos x^\circ - 6 \sin x^\circ$ 1

(ii) the value of x for which it occurs where $0 \leq x < 360$. 2

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TRIGONOMETRY

(a) Express $5\cos x - 9\sin x$ in the form $k\cos(x+a)$ where $k > 0$ and $0 < a < 2\pi$.

4

(b) Hence solve $5\cos x - 9\sin x = 7$ for $0 \leq x < 2\pi$.

3

Solve the equation $2\sin 2x^\circ - \sin^2 x^\circ = 0$, $0 \leq x < 360$.

5

Solve the equation $\sin x^\circ + 2 = 3\cos 2x^\circ$ for $0 \leq x < 360$.

5

The blades of a wind turbine are turning at a steady rate.

The height, h metres, of the tip of one of the blades above the ground at time, t seconds, is given by the formula

$$h = 36\sin(1.5t) - 15\cos(1.5t) + 65.$$

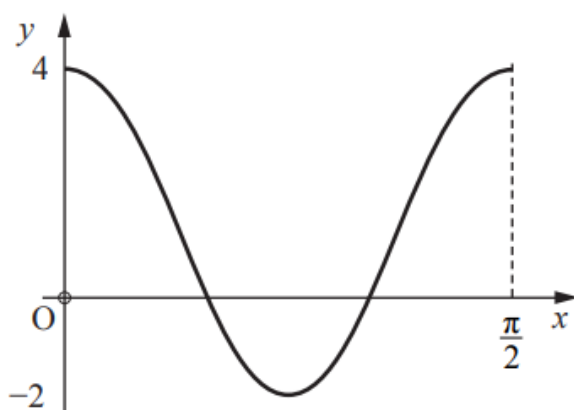
Express $36\sin(1.5t) - 15\cos(1.5t)$ in the form

$$k\sin(1.5t - a), \text{ where } k > 0 \text{ and } 0 < a < \frac{\pi}{2},$$

and hence find the **two** values of t for which the tip of this blade is at a height of 100 metres above the ground during the first turn.

8

4. The diagram shows part of the graph of the function $y = p\cos qx + r$.



Write down the values of p , q and r .

3

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TRIGONOMETRY

(a) Express $2\cos x^\circ - \sin x^\circ$ in the form $k\cos(x-a)^\circ$, $k > 0$, $0 < a < 360$.

4

(b) Hence, or otherwise, find

(i) the minimum value of $6\cos x^\circ - 3\sin x^\circ$ and

1

(ii) the value of x for which it occurs where $0 \leq x < 360$.

2

Solve $3\sin 2x^\circ + 4\cos x^\circ = 0$ for $0 \leq x < 360$.

4

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DIFFERENTIATION

Differentiate $y = \sqrt{x^3} - 2x^{-1}$, where $x > 0$. 3

Given that $y = x^{\frac{5}{3}} - \frac{10}{x^4}$, where $x \neq 0$, find $\frac{dy}{dx}$. 3

Calculate the rate of change of $d(t) = \frac{1}{2t}$, $t \neq 0$, when $t = 5$. 3

Given $h(x) = 3 \cos 2x$, find the value of $h'\left(\frac{\pi}{6}\right)$. 3

Given that $f(x) = 4 \sin\left(3x - \frac{\pi}{3}\right)$, evaluate $f'\left(\frac{\pi}{6}\right)$. 3

Given that $y = (5x^2 + 3)^7$, find $\frac{dy}{dx}$. 2

Given that $y = \frac{1}{(1-3x)^5}$, $x \neq \frac{1}{3}$, find $\frac{dy}{dx}$. 3

The function f is given by $f(x) = 12\sqrt[3]{x}$, $x > 0$.

When $x = a$ the rate of change of f with respect to x is 1.

Determine the value of a . 4

A function, f , is defined by $f(x) = (3 - 2x)^4$, where $x \in \mathbb{R}$.

Calculate the rate of change of f when $x = 4$. 3

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DIFFERENTIATION

A function, f , is defined on the set of real numbers by $f(x) = x^3 - 7x - 6$.

Determine whether f is increasing or decreasing when $x = 2$.

3

Determine the range of values of x for which the function $f(x) = 2x^3 + 9x^2 - 24x + 6$ is strictly decreasing.

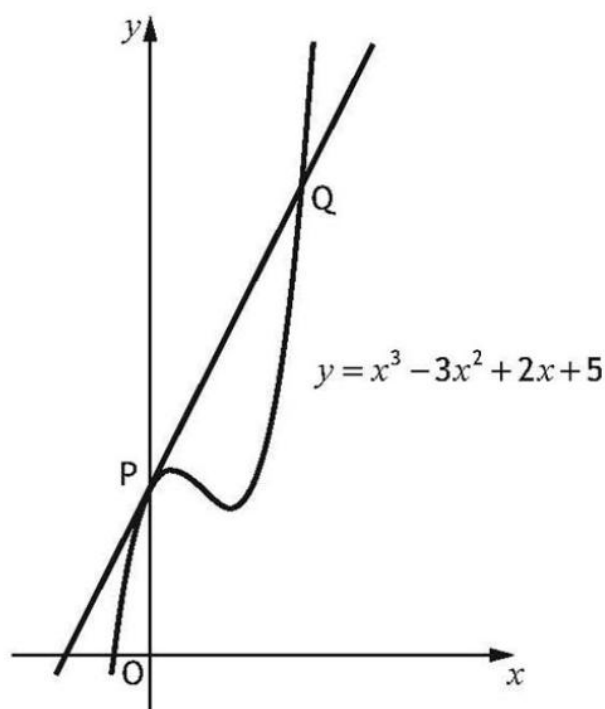
4

A curve has equation $y = x^3 - 2x^2 + 5$.

Find the equation of the tangent to this curve at the point where $x = 2$.

4

The curve with equation $y = x^3 - 3x^2 + 2x + 5$ is shown on the diagram.



(a) Write down the coordinates of P, the point where the curve crosses the y -axis.

1

(b) Determine the equation of the tangent to the curve at P.

3

(c) Find the coordinates of Q, the point where this tangent meets the curve again.

4

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DIFFERENTIATION

A function, f , is defined on $\mathbb{R}$, the set of real numbers, by $f(x) = x^3 + 3x^2 - 9x + 5$.

Find the coordinates of the stationary points of f and determine their nature.

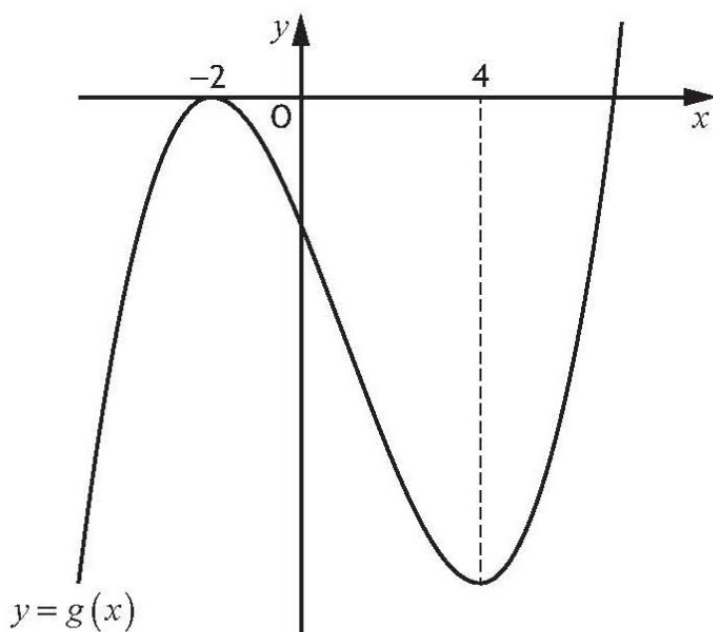
6

Find the x -coordinates of the stationary points on the curve with equation

$$y = \frac{1}{2}x^4 - 2x^3 + 6.$$

4

The diagram below shows the graph of a cubic function $y = g(x)$, with stationary points at $x = -2$ and $x = 4$.



On the diagram in your answer booklet, sketch the graph of $y = g'(x)$.

2

A sector with a particular fixed area has radius x cm.

The perimeter, P cm, of the sector is given by

$$P = 2x + \frac{128}{x}.$$

Find the minimum value of P .

6

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DIFFERENTIATION

- (a) Determine the coordinates of the stationary points on the curve with equation

$$y = \frac{1}{3}x^3 - x^2 - 3x + 1.$$

4

- (b) Hence, determine the greatest and least values of y in the interval $-1 \leq x \leq 6$.

2

- (a) Find the x -coordinate of the stationary point on the curve

$$\text{with equation } y = 6x - 2\sqrt{x^3}.$$

4

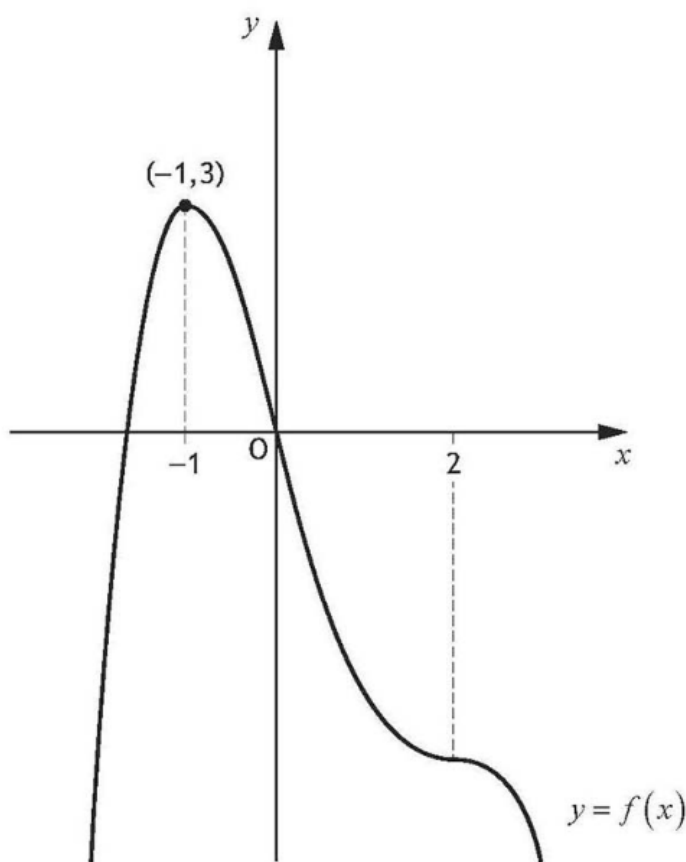
- (b) Hence, determine the greatest and least values of y in the interval $1 \leq x \leq 9$.

3

The diagram shows the graph of a quartic function $y = f(x)$.

A maximum turning point occurs at $(-1, 3)$.

The graph of $y = f(x)$ also has a point of inflection at $x = 2$.



- (a) Determine the coordinates of the maximum turning point on the graph of $y = f(x-4) + 2$.

2

- (b) On the diagram in your answer booklet, sketch the graph of $y = f'(x)$.

3

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DIFFERENTIATION

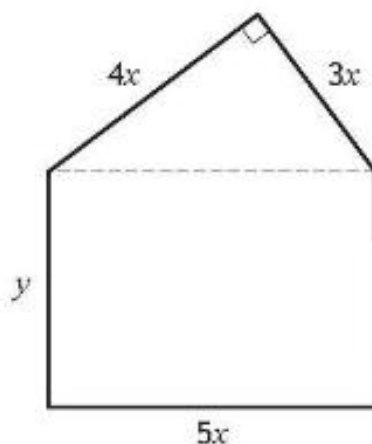
A hotel owner is designing signs showing the room numbers.



Each sign is a rectangle with a right-angled triangle above it.

The length and breadth of the rectangle are $5x$ centimetres and y centimetres respectively.

The shorter sides of the triangle are $3x$ centimetres and $4x$ centimetres.



The area of the sign is 150 square centimetres.

(a) Show that the perimeter, P cm, of the sign is given by

$$P = 9.6x + \frac{60}{x}.$$

Each sign will be lit using a lighting strip placed around its perimeter.

The hotel owner requires the perimeter, P , of the sign to be as small as possible.

(b) Find the minimum value of P .

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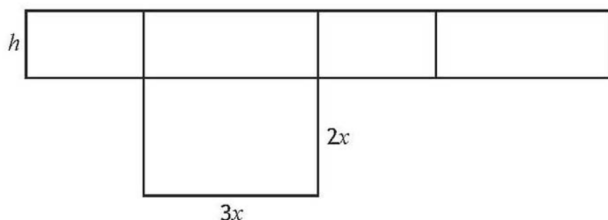
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DIFFERENTIATION

A net of an open box is shown.

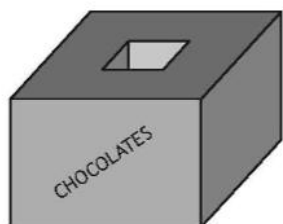
The box is a cuboid with height h centimetres.

The base is a rectangle measuring $3x$ centimetres by $2x$ centimetres.



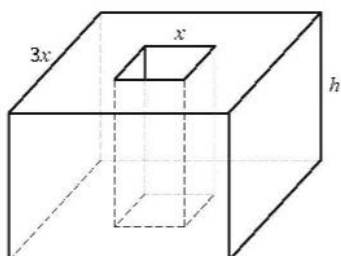
- (a) (i) Express the area of the net, $A \text{ cm}^2$, in terms of h and x . 1
- (ii) Given that $A = 7200 \text{ cm}^2$, show that the volume of the box, $V \text{ cm}^3$, is given by $V = 4320x - \frac{18}{5}x^3$. 2
- (b) Determine the value of x that maximises the volume of the box. 4

A manufacturer of chocolates is launching a new product in novelty shaped cardboard boxes.



The box is a cuboid with a cuboid shaped tunnel through it.

- The height of the box is h centimetres
- The top of the box is a square of side $3x$ centimetres
- The end of the tunnel is a square of side x centimetres
- The volume of the box is 2000 cm^3



- (a) Show that the total surface area, $A \text{ cm}^2$, of the box is given by

$$A = 16x^2 + \frac{4000}{x}$$

3

- (b) To minimise the cost of production, the surface area, A , of the box should be as small as possible.

Find the minimum value of A .

6

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EXPONENTIALS & LOGS

Express $\log_a 5 + \log_a 80 - 2\log_a 10$ in the form $\log_a k$ where k is a positive integer. 3

Evaluate $3\log_3 2 + \log_3 \frac{1}{24}$. 3

Evaluate $\log_2 5 + \log_2 \frac{1}{40}$. 2

Given that $a \in \mathbb{R}$ and that $\log_8 a$ is negative, state the range of possible values of a . 1

Given that $\log_a 75 = 2 + \log_a 3$, $a > 0$, find the value of a . 3

Solve $\log_6 x + \log_6 (x+5) = 2$, where $x > 0$. 4

The number of electric vehicles worldwide can be modelled by

$$N = 6.8e^{kt}$$

where:

- N is the estimated number of vehicles in millions
- t is the number of years since the end of 2020
- k is a constant.

(a) Use the model to estimate the number of electric vehicles worldwide at the end of 2020. 1

At the end of 2030, it is estimated there will be 125 million electric vehicles worldwide.

(b) Determine the value of k . 4

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EXPONENTIALS & LOGS

A radioactive substance, which has been collected, decays over time.

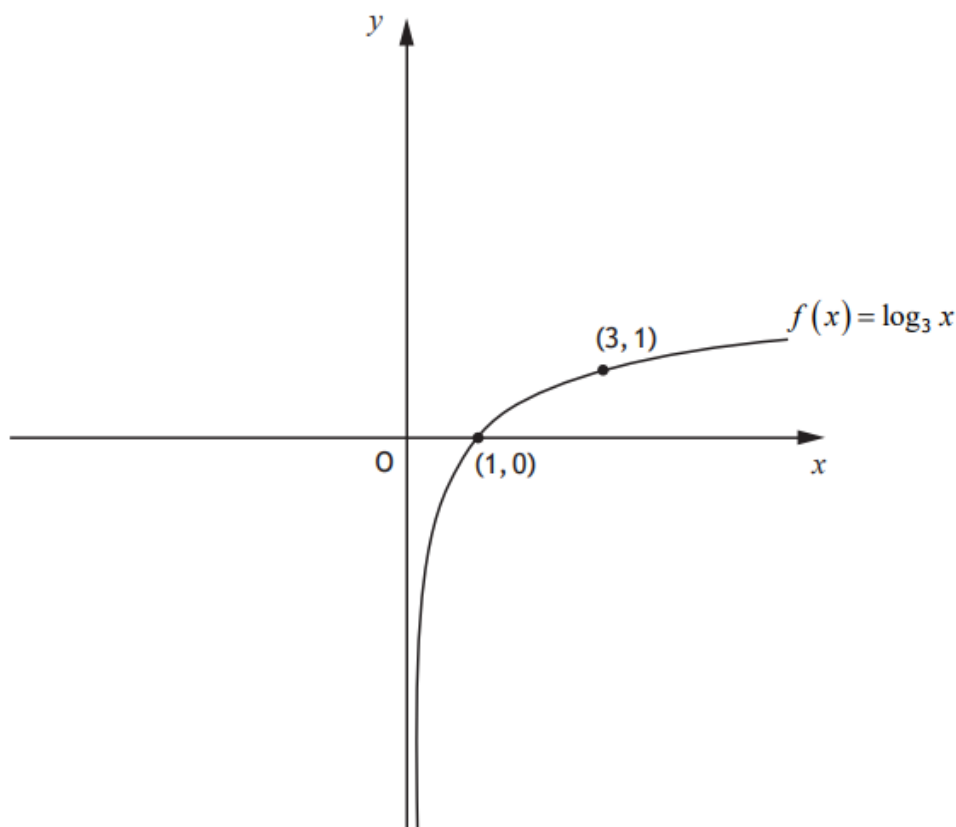
The mass of the radioactive substance remaining is modelled by

$$M = 150e^{-0.0054t}$$

where M is the mass, in micrograms, t years after the radioactive substance was collected.

- (a) Determine the initial mass of the radioactive substance. 1
- (b) Calculate the time taken for the mass of the radioactive substance to decay to 120 micrograms. 4

9. The diagram shows the graph of the function $f(x) = \log_3 x$, where $x > 0$.



The inverse function, f^{-1} , exists.

On the diagram in your answer booklet, sketch the graph of $y = f^{-1}(x) - 1$.

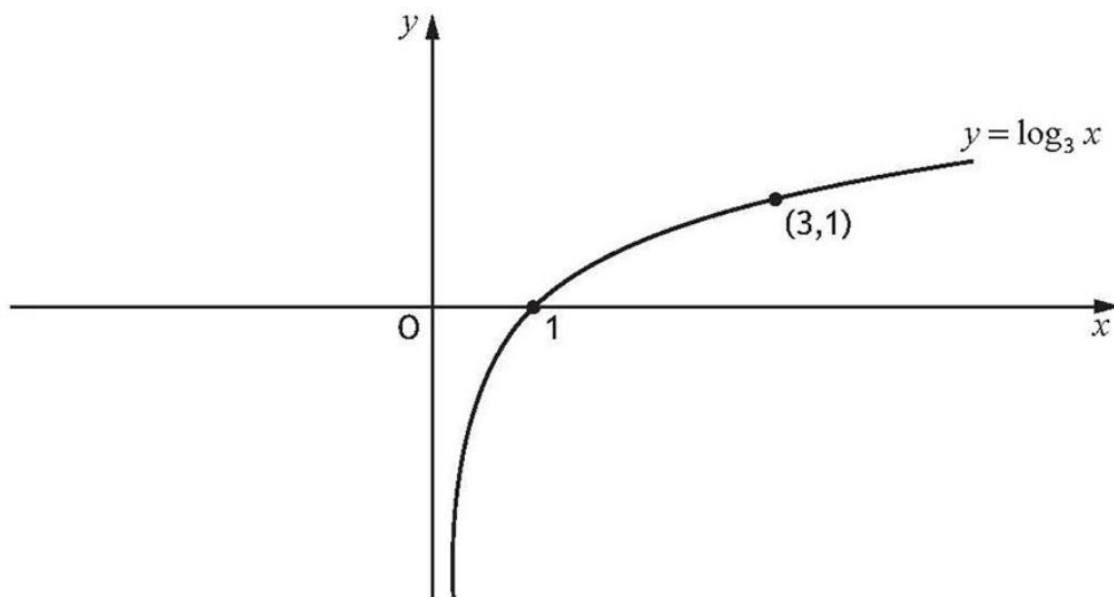
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EXPONENTIALS & LOGS

The diagram shows the curve with equation $y = \log_3 x$.



- On the diagram in your answer booklet, sketch the curve with equation $y = 1 - \log_3 x$. 2
- Determine the exact value of the x -coordinate of the point of intersection of the two curves. 3

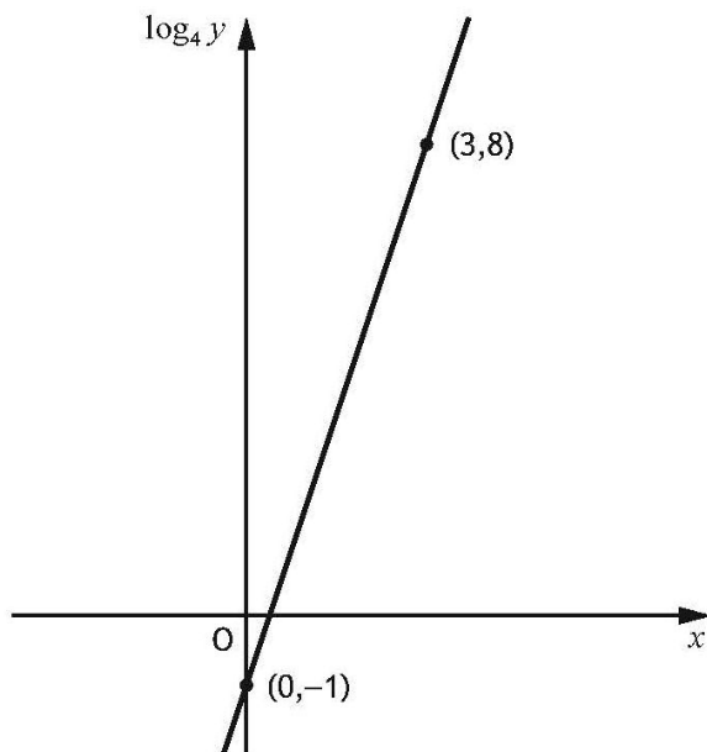
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EXPONENTIALS & LOGS

Two variables, x and y , are connected by the equation $y = ab^x$.

The graph of $\log_4 y$ against x is a straight line as shown.



Find the values of a and b .

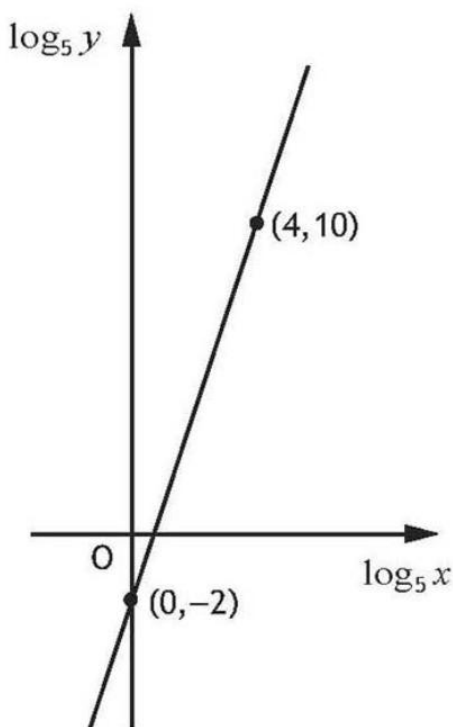
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EXPONENTIALS & LOGS

Two variables, x and y , are connected by the equation $y = ax^b$.

The graph of $\log_5 y$ against $\log_5 x$ is a straight line as shown.



Find the values of a and b .

5

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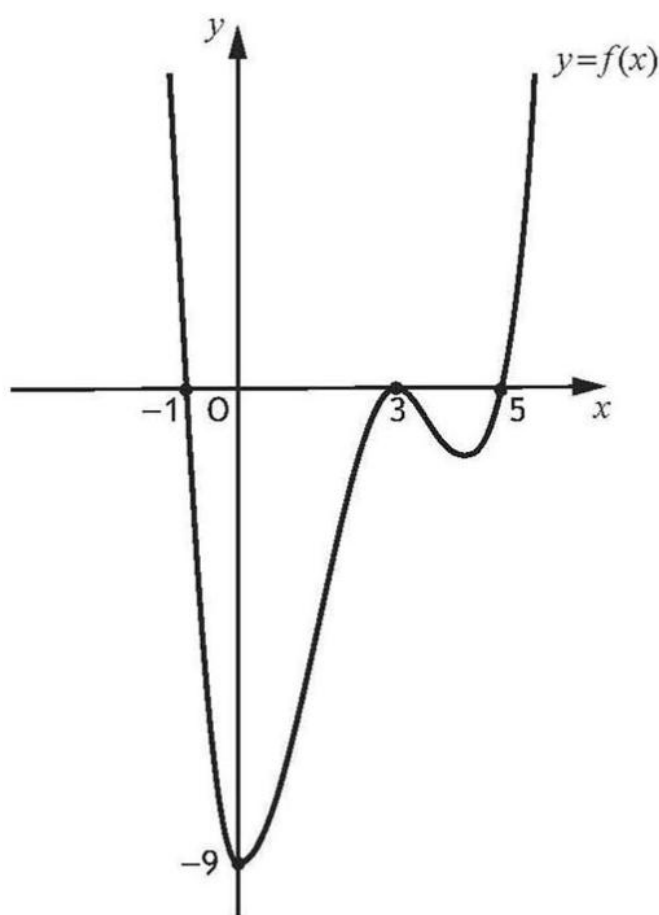
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QUADRATICS & POLYNOMIALS

- (a) Show that $(x-1)$ is a factor of $2x^4 + 3x^3 - 4x^2 - 3x + 2$. 2
- (b) Hence, or otherwise, factorise $2x^4 + 3x^3 - 4x^2 - 3x + 2$ fully. 4

- (a) Show that $(x+3)$ is a factor of $5x^3 + 16x^2 - x - 12$. 2
- (b) Hence, or otherwise, solve $5x^3 + 16x^2 - x - 12 = 0$. 3

The diagram shows the graph of $y = f(x)$, where $f(x)$ is a quartic function.



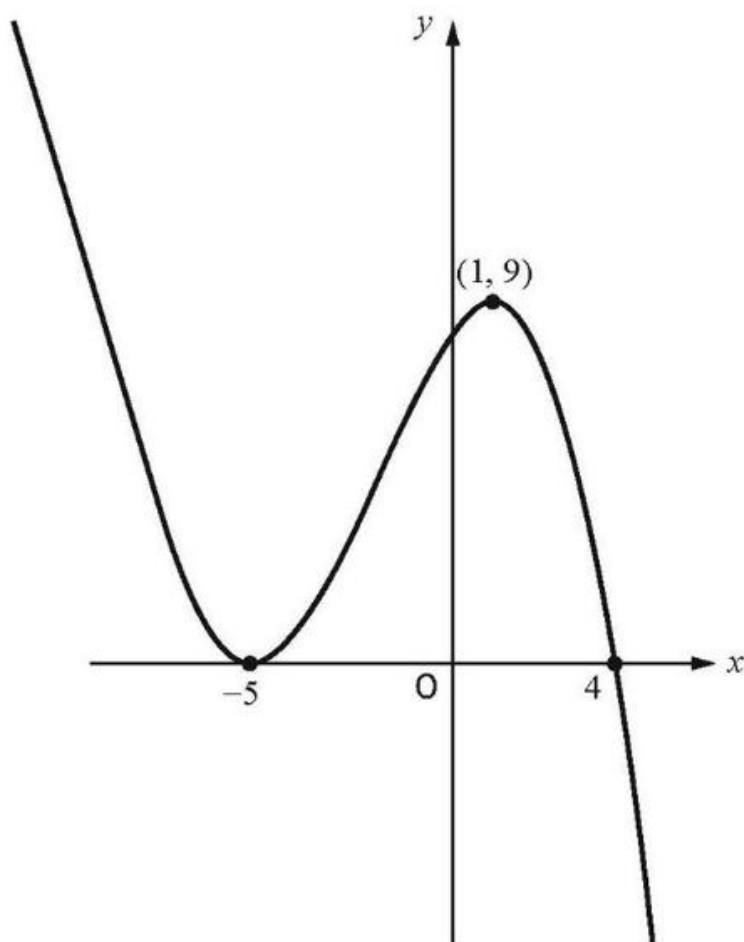
Express $f(x)$ in the form $f(x) = k(x+a)^2(x+b)(x+c)$. 3

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QUADRATICS & POLYNOMIALS

The diagram below shows the graph with equation $y = f(x)$, where $f(x) = k(x-a)(x-b)^2$.



(a) Find the values of a , b and k .

3

(b) For the function $g(x) = f(x) - d$, where d is positive, determine the range of values of d for which $g(x)$ has exactly one real root.

1

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QUADRATICS & POLYNOMIALS

A function, f , is defined on the set of real numbers.

The derivative of f is $f'(x) = (x+5)(2-x)$.

- (a) Find the x -coordinates of the stationary points on the curve with equation $y = f(x)$ and determine their nature.

3

It is known that:

- f is a cubic function
- $f(0) < 0$
- the equation $f(x) = 0$ has exactly one solution. The solution lies between -10 and 10 .

- (b) Draw a sketch of a possible graph of $y = f(x)$ on the diagram in your answer booklet.

3

4. Express $-3x^2 - 6x + 7$ in the form $a(x+b)^2 + c$.

3

7. (a) Express $-6x^2 + 24x - 25$ in the form $p(x+q)^2 + r$.

3

- (b) Given that $f(x) = -2x^3 + 12x^2 - 25x + 9$,
show that $f(x)$ is strictly decreasing for all $x \in \mathbb{R}$.

3

2. Functions f and g are defined on suitable domains by

$$f(x) = 10 + x \text{ and } g(x) = (1+x)(3-x) + 2.$$

- (a) Find an expression for $f(g(x))$.

2

- (b) Express $f(g(x))$ in the form $p(x+q)^2 + r$.

3

- (c) Another function h is given by $h(x) = \frac{1}{f(g(x))}$.

What values of x cannot be in the domain of h ?

2

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QUADRATICS & POLYNOMIALS

The equation $x^2 + (m-4)x + (2m-3) = 0$ has no real roots.

Determine the range of values for m .

4

Justify your answer.

The equation $9x^2 + 3kx + k = 0$ has two real and distinct roots.

Determine the range of values for k .

Justify your answer.

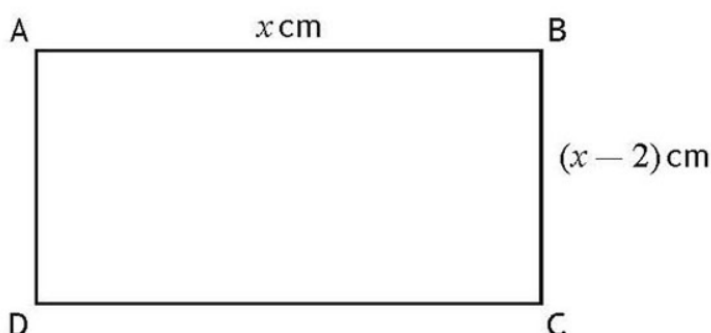
4

The equation $x^2 + (m-3)x + m = 0$ has two real and distinct roots.

Determine the range of values for m .

4

ABCD is a rectangle with sides of lengths x centimetres and $(x-2)$ centimetres, as shown.



If the area of ABCD is less than 15 cm^2 , determine the range of possible values of x .

4

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CIRCLES

Circle C_1 has equation $x^2 + y^2 - 6x - 2y - 26 = 0$.

Circle C_2 has centre $(4, -2)$.

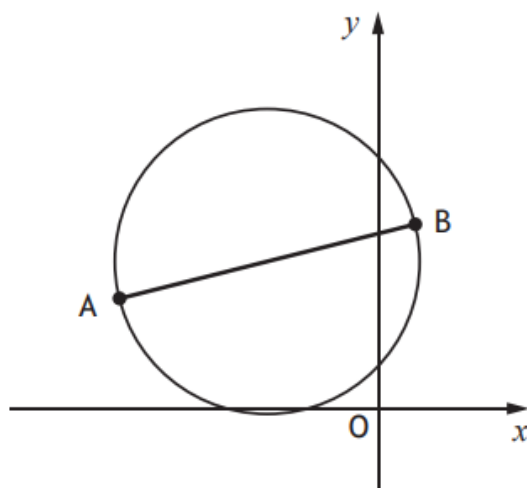
The radius of C_2 is equal to the radius of C_1 .

Find the equation of circle C_2 .

2

4. A and B are the points $(-7, 3)$ and $(1, 5)$.

AB is a diameter of a circle.



Find the equation of this circle.

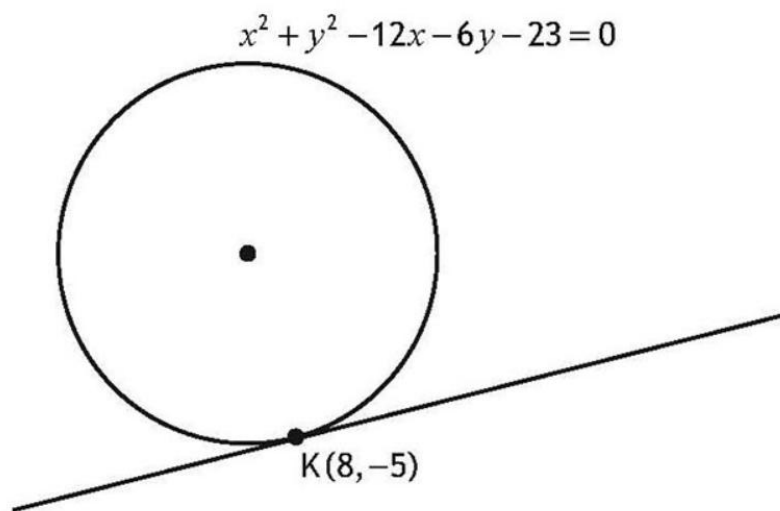
3

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CIRCLES

The point $K(8, -5)$ lies on the circle with equation $x^2 + y^2 - 12x - 6y - 23 = 0$.



Find the equation of the tangent to the circle at K .

4

$T(-2, -5)$ lies on the circumference of the circle with equation

$$(x + 8)^2 + (y + 2)^2 = 45.$$

(a) Find the equation of the tangent to the circle passing through T .

4

(b) This tangent is also a tangent to a parabola with equation $y = -2x^2 + px + 1 - p$, where $p > 3$.

Determine the value of p .

6

The line $y = 2x$ is a tangent to the circle with equation $x^2 + y^2 - 14x - 8y + 45 = 0$.

Determine the coordinates of the point of contact.

4

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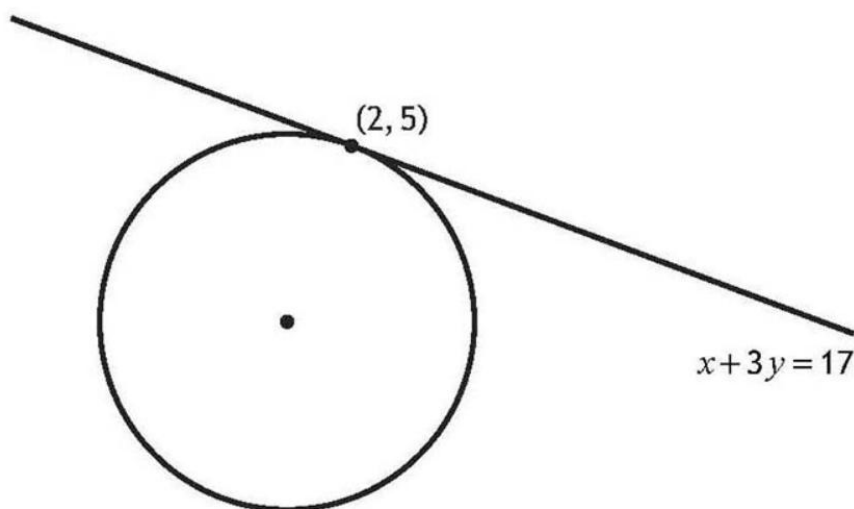
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CIRCLES

Find the coordinates of the points of intersection of the line with equation $y = x + 1$ and the circle with equation $x^2 + y^2 - 2x + 6y - 15 = 0$.

4

The line $x + 3y = 17$ is a tangent to a circle at the point $(2, 5)$.



The centre of the circle lies on the y -axis.

Find the coordinates of the centre of the circle.

4

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CIRCLES

Show that the line with equation $y = 3x - 5$ is a tangent to the circle with equation $x^2 + y^2 + 2x - 4y - 5 = 0$ and find the coordinates of the point of contact.

5

P and Q are the points (4, 10) and (6, 2) respectively.

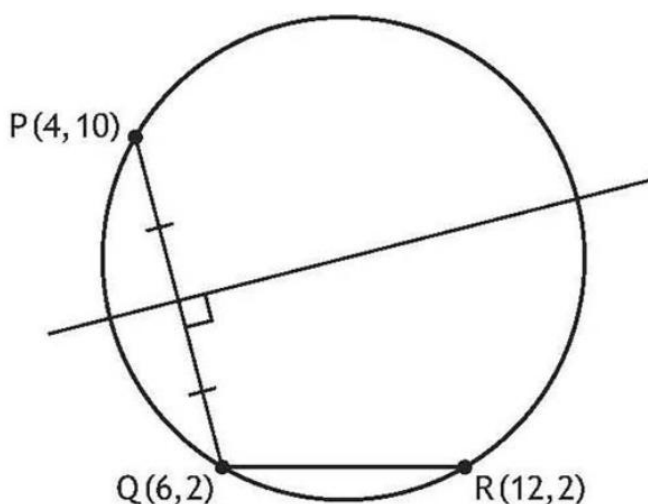
(a) Find the equation of the perpendicular bisector of PQ.

4

The point R has coordinates (12, 2).

A circle passes through the points P, Q and R.

The chord QR is horizontal.



(b) Find the equation of the circle.

4

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CIRCLES

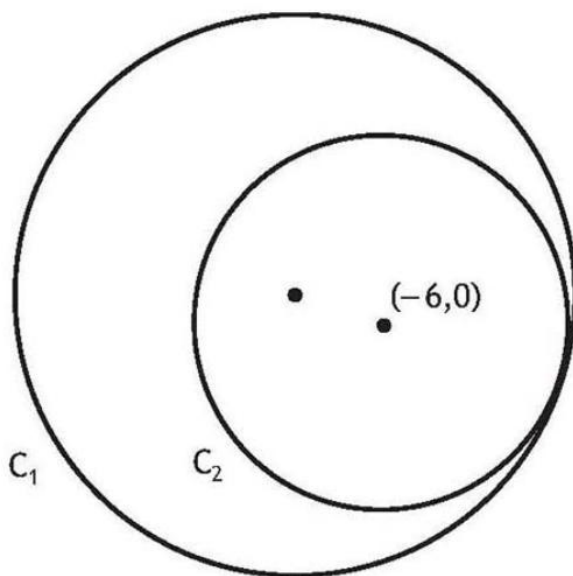
The circle C_1 has equation $x^2 + y^2 + 18x - 2y - 8 = 0$.

(a) Find the centre and radius of C_1 .

2

A second circle, C_2 , touches C_1 internally.

The centre of C_2 is $(-6, 0)$.



(b) Determine the equation of C_2 .

2

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CIRCLES

Circle C_1 has equation $(x+5)^2 + (y-6)^2 = 9$.

(a) State the centre and radius of C_1 .

2

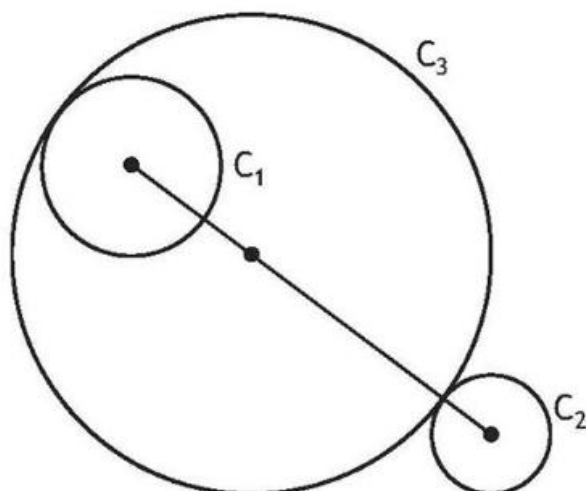
Circle C_2 has equation $x^2 + y^2 - 14x + 6y + 54 = 0$.

(b) State the centre and radius of C_2 .

2

Circles C_1 , C_2 and C_3 are touching as shown in the diagram.

The centre of circle C_3 lies on the line joining the centres of C_1 and C_2 .



(c) Determine the equation of C_3 .

3

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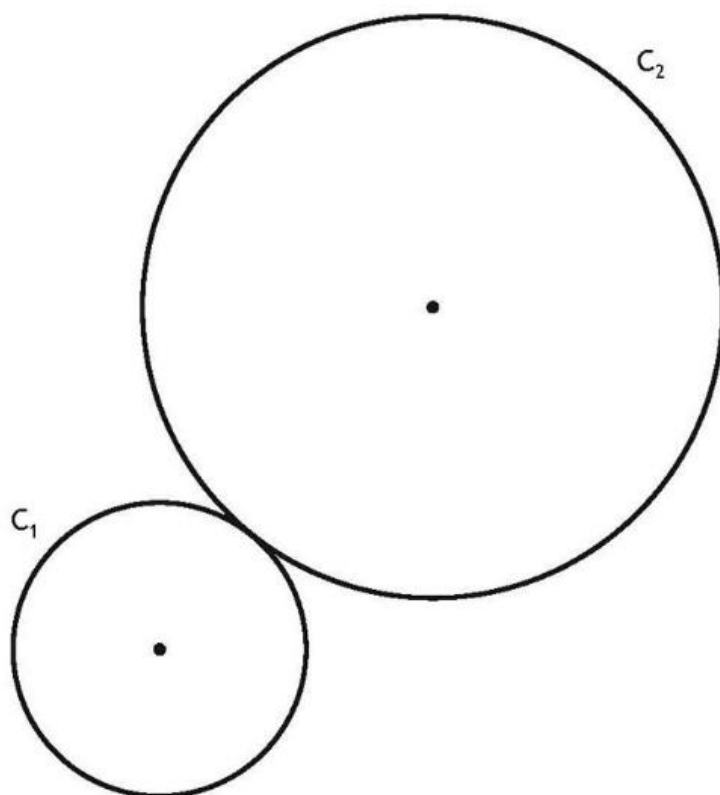
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CIRCLES

Circle C_1 has equation $x^2 + y^2 + 6x + 10y + 9 = 0$.

The centre of circle C_2 is $(9, 11)$.

Circles C_1 and C_2 touch externally.



(a) Determine the radius of C_2 .

4

A third circle, C_3 , is drawn such that:

- both C_1 and C_2 touch C_3 internally
- the centres of C_1 , C_2 and C_3 are collinear.

(b) Determine the equation of C_3 .

4

The point P has coordinates $(4, k)$.

C is the centre of the circle with equation $(x-1)^2 + (y+2)^2 = 25$.

(a) Show that the distance between the points P and C is given by $\sqrt{k^2 + 4k + 13}$.

2

(b) Hence, or otherwise, find the range of values of k such that P lies outside the circle.

4

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INTEGRATION

Find $\int \left(\frac{12}{x^2} + x^{\frac{1}{2}} \right) dx, x > 0.$

4

Find $\int (2x^5 - 6\sqrt{x}) dx, x \geq 0.$

4

Evaluate $\int_0^{\frac{\pi}{7}} \sin 5x \, dx.$

3

Find $\int 7 \cos \left(4x + \frac{\pi}{3} \right) dx.$

2

Find $\int (3x+2)^7 \, dx.$

2

Evaluate $\int_0^{\frac{\pi}{9}} \cos \left(3x - \frac{\pi}{6} \right) dx.$

4

Evaluate $\int_{-4}^9 \frac{1}{\sqrt[3]{(2x+9)^2}} \, dx.$

5

For a function, f , defined on the set of real numbers, $\mathbb{R}$, it is known that

- the rate of change of f with respect to x is given by $3x^2 - 16x + 11$
- the graph with equation $y = f(x)$ crosses the x -axis at $(7,0)$.

Express $f(x)$ in terms of x .

5

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INTEGRATION

Given that:

- $\frac{dy}{dx} = 6 \cos x + 8 \sin 2x$, and
- $y = 4$ when $x = \frac{\pi}{6}$,

express y in terms of x .

4

The rate of change of the temperature, T °C of a mug of coffee is given by

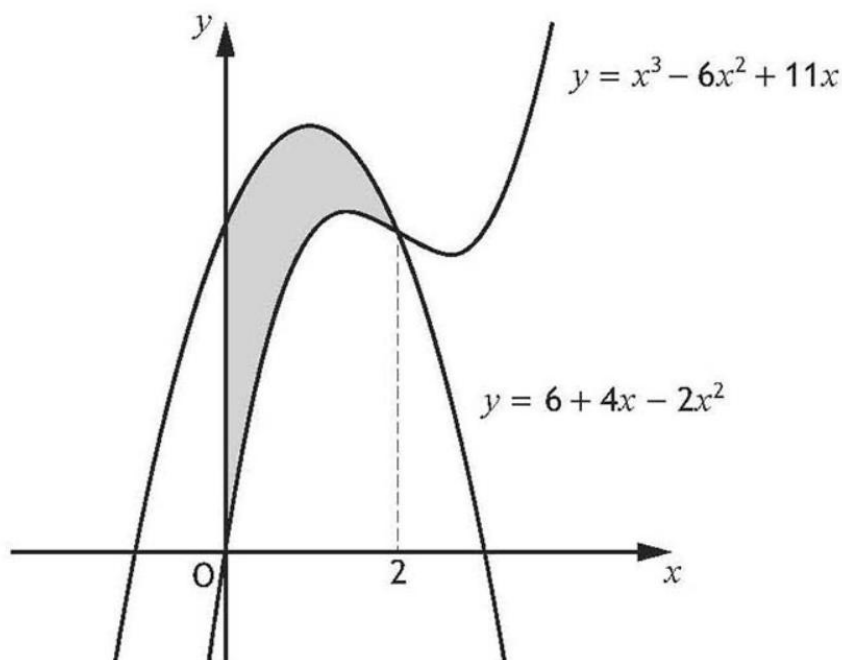
$$\frac{dT}{dt} = \frac{1}{25}t - k, \quad 0 \leq t \leq 50$$

- t is the elapsed time, in minutes, after the coffee is poured into the mug
- k is a constant
- initially, the temperature of the coffee is 100 °C
- 10 minutes later the temperature has fallen to 82 °C.

Express T in terms of t .

6

The diagram shows the curve with equation $y = x^3 - 6x^2 + 11x$ intersecting the curve with equation $y = 6 + 4x - 2x^2$ at $x = 2$.



Calculate the shaded area.

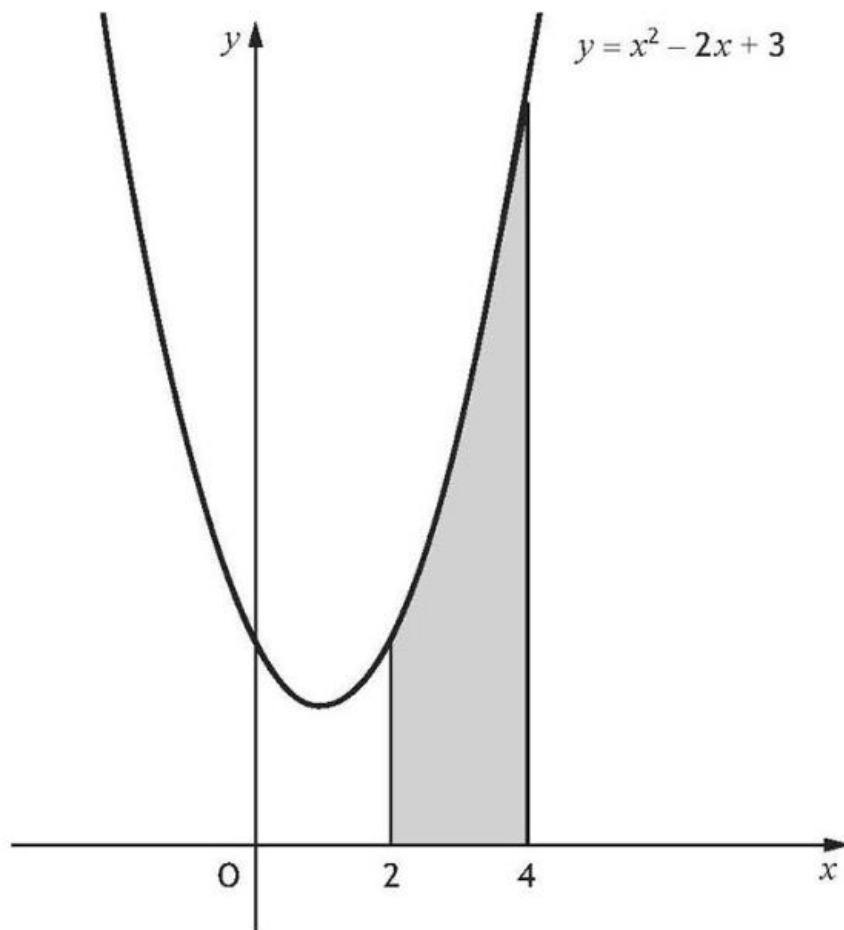
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INTEGRATION

The diagram shows the graph of $y = x^2 - 2x + 3$.



Calculate the shaded area.

4

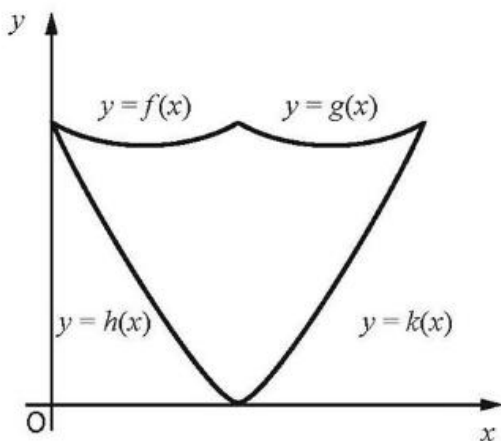
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INTEGRATION

A wall plaque is to be made to commemorate the 150th anniversary of the publication of "*Alice's Adventures in Wonderland*".

The edges of the wall plaque can be modelled by parts of the graphs of four quadratic functions as shown in the sketch.



- $f(x) = \frac{1}{4}x^2 - \frac{1}{2}x + 3$
- $g(x) = \frac{1}{4}x^2 - \frac{3}{2}x + 5$
- $h(x) = \frac{3}{8}x^2 - \frac{9}{4}x + 3$
- $k(x) = \frac{3}{8}x^2 - \frac{3}{4}x$

(a) Find the x -coordinate of the point of intersection of the graphs with equations $y = f(x)$ and $y = g(x)$.

2

The graphs of the functions $f(x)$ and $h(x)$ intersect on the y -axis.

The plaque has a vertical line of symmetry.

(b) Calculate the area of the wall plaque.

7

(a) Given that $y = (x^2 + 7)^{\frac{1}{2}}$, find $\frac{dy}{dx}$.

2

(b) Hence find $\int \frac{4x}{\sqrt{x^2 + 7}} dx$.

1

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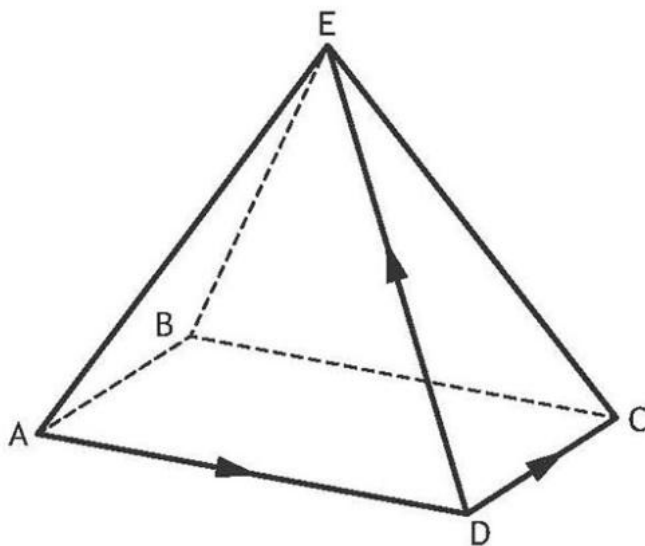
VECTORS

E,ABCD is a rectangular-based pyramid as shown.

$$\overrightarrow{AD} = 6\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$$

$$\overrightarrow{DC} = 2\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$$

$$\overrightarrow{DE} = -4\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$$



Express $\overrightarrow{BE}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$.

2

Vectors $\mathbf{a}$ and $\mathbf{b}$ are such that $\mathbf{a} = 4\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = -2\mathbf{i} + \mathbf{j} + p\mathbf{k}$.

(a) Express $2\mathbf{a} + \mathbf{b}$ in component form.

1

(b) Hence find the values of p for which $|2\mathbf{a} + \mathbf{b}| = 7$.

3

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VECTORS

Three vectors can be expressed as follows:

$$\vec{FG} = -2\mathbf{i} - 6\mathbf{j} + 3\mathbf{k}$$

$$\vec{GH} = 3\mathbf{i} + 9\mathbf{j} - 7\mathbf{k}$$

$$\vec{EH} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$$

(a) Find $\vec{FH}$.

2

(b) Hence, or otherwise, find $\vec{FE}$.

2

(a) Show that the points A $(-3, 2, -1)$, B $(6, -1, 5)$ and C $(12, -3, 9)$ are collinear.

3

(b) State the ratio in which B divides AC.

1

P and Q have coordinates $(-6, 1, 2)$ and $(-1, 11, -8)$ respectively.

Find the coordinates of the point R which divides PQ in the ratio 2:3.

2

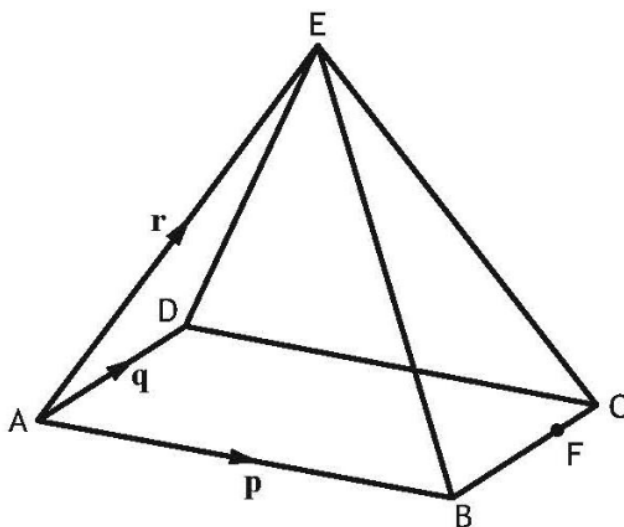
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VECTORS

E,ABCD is a rectangular based pyramid.

$\vec{AB} = \mathbf{p}$, $\vec{AD} = \mathbf{q}$ and $\vec{AE} = \mathbf{r}$.



(a) Express $\vec{BE}$ in terms of $\mathbf{p}$ and $\mathbf{r}$.

1

Point F divides BC in the ratio 3:1.

(b) Express vector $\vec{EF}$ in terms of $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$.

2

Vectors $\mathbf{u}$ and $\mathbf{v}$ have components $\begin{pmatrix} p \\ -2 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 2p+16 \\ -3 \\ 6 \end{pmatrix}$, $p \in \mathbb{R}$.

(a) (i) Find an expression for $\mathbf{u} \cdot \mathbf{v}$.

1

(ii) Determine the values of p for which $\mathbf{u}$ and $\mathbf{v}$ are perpendicular.

3

(b) Determine the value of p for which $\mathbf{u}$ and $\mathbf{v}$ are parallel.

2

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VECTORS

The coordinates of points D, E and F are given by D (2, -3, 4), E (1, 1, -2) and F (3, 2, 1).

(a) Express $\vec{ED}$ and $\vec{EF}$ in component form.

(b) (i) Calculate $\vec{ED} \cdot \vec{EF}$.

(ii) Hence, or otherwise, calculate the size of angle DEF.

Vectors $\mathbf{u}$ and $\mathbf{v}$ are defined by $\mathbf{u} = \begin{pmatrix} -1 \\ 4 \\ -3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -7 \\ 8 \\ 5 \end{pmatrix}$.

(a) Find $\mathbf{u} \cdot \mathbf{v}$.

(b) Calculate the acute angle between $\mathbf{u}$ and $\mathbf{v}$.

The vectors $\mathbf{u}$ and $\mathbf{v}$ are such that:

- $\mathbf{u} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

- $\mathbf{v} = \begin{pmatrix} 1 \\ 3 \\ k \end{pmatrix}$

- the angle between $\mathbf{u}$ and $\mathbf{v}$ is 45° .

Find the value of k , where $k > 0$.

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